UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Computing

SOFTWARE REQUIREMENTS SPECIFICATION

UTM BORROW

Campus Resource Sharing & Circular Economy Platform

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Reviewed By	Faculty Supervisor / Project Dr.

This document is prepared in accordance with IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.

1. INTRODUCTION

1.1 Purpose of This Document

This Software Requirements Specification (SRS) document formally defines the complete functional and non-functional requirements for UTM Borrow — a web and mobile-based campus resource sharing platform developed for Universiti Teknologi Malaysia (UTM). This document is intended for use by the software development team, the faculty supervisor, university IT administrators, and any future maintenance engineers. It serves as the primary contractual reference between stakeholders and developers, specifying what the system must do, how it must perform, and the constraints within which it must operate.

This document follows the IEEE Std 830-1998 standard structure and will be versioned throughout the software development lifecycle (SDLC).

1.2 Project Overview

UTM Borrow is an authenticated, closed-ecosystem digital platform designed to facilitate the peer-to-peer (P2P) borrowing and lending of physical academic resources within the UTM campus community. The platform addresses a critical financial inefficiency: students routinely purchase expensive tools, lab equipment, textbooks, and stationery for single-module use, then store them indefinitely.

The system functions as an intelligent, real-time inventory management and handover-secured transaction engine — not a simple classified ads board. It enforces identity verification, tracks item custody state, manages borrow lifecycles, and generates accountability records. In future phases, it will incorporate a merit-reward system and a symbolic maintenance fee model to ensure long-term sustainability.

1.3 Scope

UTM Borrow is scoped to serve the following user populations exclusively within the UTM ecosystem:

- Undergraduate and postgraduate students enrolled at UTM
- UTM academic staff and faculty members
- UTM administrative officers (as platform moderators)

The platform is out-of-scope for:

- General public or non-UTM users
- Cross-university item sharing in Phase 1
- Commercial resale or auction features
- Physical delivery or courier logistics (all handovers are in-person)

1.4 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
UTM	Universiti Teknologi Malaysia

P2P	Peer-to-Peer — direct transaction between two users without intermediary
OAuth2	Open Authorization 2.0 — industry-standard protocol for authentication delegation
QR Code	Quick Response code used as a cryptographic handover token in this system
Lender	A UTM user who registers an item and approves a borrow request
Borrower	A UTM user who requests to borrow an item from a Lender
Lease Cycle	The full lifecycle of a single borrow transaction from request to return
Merit Points	Gamified reward currency awarded for lending activity (Phase 2+)
Good Standing	A user account status indicating a trust rating above 3.0/5.0
Cloud Function	Serverless backend logic triggered by database events (Firebase)
Firestore	Google's NoSQL cloud-hosted document database
API	Application Programming Interface
UTMCS	UTM Campus Sustainability — the university sustainability office
BI	Business Intelligence — data analytics and visualization
SDLC	Software Development Lifecycle
RBAC	Role-Based Access Control
JWT	JSON Web Token — secure authentication token format

1.5 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- UTM ICT Policy and Data Governance Framework (Internal Document)
- Google Firebase Documentation — Firestore, Authentication, Cloud Functions
- OAuth 2.0 Authorization Framework — RFC 6749 (IETF)
- UTM Smart Campus Vision 2025 Strategic Plan
- OWASP Top 10 Web Application Security Risks (2023 Edition)

1.6 Document Overview

This SRS is organized into six major sections. Section 1 provides introductory and contextual information. Section 2 describes the overall system description including product perspective, user classes, and assumptions. Section 3 defines all functional requirements organized by subsystem. Section 4 defines all non-functional requirements. Section 5 provides system architecture and data

model references. Section 6 covers appendices including use case tables and glossary supplements.

2. OVERALL SYSTEM DESCRIPTION

2.1 Product Perspective

UTM Borrow is a standalone, purpose-built platform that integrates with the existing UTM student and staff identity infrastructure through OAuth2-based Single Sign-On (SSO). It does not replace any existing UTM system but complements the university's digital services ecosystem.

The system can be visualized as three interconnected layers:

- Identity Layer: UTM OAuth2 SSO integration for verified, non-anonymous accounts
- Application Layer: React Native (mobile) and React.js (web) frontend applications
- Backend Layer: Node.js REST API with Firebase Firestore database and Cloud Functions

The platform has no physical infrastructure requirements beyond standard web hosting and uses Firebase's scalable cloud infrastructure. All item handovers are physical and in-person; the platform facilitates the coordination, verification, and recording of these handovers digitally.

2.2 Product Functions — High-Level Summary

Function Group	Description	Phase
Identity & Authentication	UTM OAuth2 SSO login, profile creation, role assignment	1
Item Listing Management	Create, edit, publish, and deactivate borrowable item listings	1
Search & Discovery	Geolocation-filtered search with category and availability filters	1
Borrow Request Flow	Send, approve, reject, and track borrow requests	1
QR Handover System	Timed cryptographic QR codes for dual-scan item handover verification	1
Lease Cycle Management	Real-time item custody state tracking throughout borrow lifecycle	1
Notification Engine	Push and email alerts for requests, deadlines, and system events	1
Condition Reporting	Pre- and post-borrow photo documentation and damage reporting	1
Trust & Rating System	5-star rating system tied to Good Standing status enforcement	1
Admin Moderation Panel	Dispute resolution, user management, and system oversight	1
Maintenance Fee Wallet	1RM symbolic transaction fee and digital wallet (Phase 2)	2
Merit Points Engine	Gamified lending rewards and university recognition system (Phase 2)	2
Analytics & BI Dashboard	Campus-wide savings and sustainability metrics for admin (Phase 2)	2

Faculty-Only Visibility	Restricted listing visibility controls for sensitive equipment (Phase 2)	2
Calendar Integration	Automatic Google Calendar event creation for borrow deadlines (Phase 3)	3
Export & Reporting API	PDF export of transaction logs for disciplinary proceedings (Phase 3)	3

2.3 User Classes and Characteristics

2.3.1 Student User

The primary user class. Students can act as both Lenders and Borrowers within the same account. They are authenticated via their UTM student portal credentials. They interact with the full item lifecycle, search, and handover subsystems.

2.3.2 Staff User

Academic and administrative staff may register items and approve borrows. Staff accounts may have elevated Trust Rating floors. Staff do not manage the platform but can set items to Faculty-Only visibility in Phase 2.

2.3.3 Platform Administrator

University-designated administrators who access the Admin Moderation Panel. Administrators can suspend accounts, resolve disputes, view all transaction logs, and export compliance reports. Administrators cannot modify transaction outcomes retroactively.

2.3.4 System (Automated Processes)

Firebase Cloud Functions acting as automated agents that trigger notifications, transition Lease Cycle states, expire timed QR tokens, and enforce return deadline penalties without human intervention.

2.4 Operating Environment

- Mobile Platform: iOS 14+ and Android 10+ via React Native application
- Web Platform: Modern browsers (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- Backend: Node.js 18 LTS on Firebase Cloud Functions (serverless)
- Database: Google Firebase Firestore (NoSQL document store)
- Authentication: UTM OAuth2 SSO endpoint (existing university infrastructure)
- Push Notifications: Firebase Cloud Messaging (FCM)
- Storage: Firebase Cloud Storage (item photos, condition reports)
- Hosting: Firebase Hosting (web frontend)

2.5 Design and Implementation Constraints

- All user accounts must be authenticated against the UTM identity provider — no self-registration with external email addresses is permitted
- Item handover can only be finalized via dual QR code scan — verbal or chat-based confirmation is not accepted by the system
- QR handover tokens must expire within 60 seconds of generation to prevent screenshot-based fraud

- All item photos must be stored securely in Firebase Cloud Storage with access controlled by user authentication tokens
- The system must comply with Malaysia's Personal Data Protection Act 2010 (PDPA 2010) for all user data handling
- The platform must support Bahasa Malaysia and English for all user-facing text in Phase 1
- API endpoints must implement RBAC to prevent unauthorized cross-user data access

2.6 Assumptions and Dependencies

- UTM's OAuth2 SSO endpoint is accessible and returns a valid matriculation number, user role (student/staff), and college/faculty affiliation
- Firebase services (Firestore, Cloud Functions, FCM, Storage, Hosting) remain available and within the project's cost budget
- Users own smartphones with camera access for QR scanning and item photo documentation
- UTM campus geolocation data (college buildings with GPS coordinates) is available for proximity filtering
- The university provides a designated email domain pattern (e.g., @graduate.utm.my, @utm.my) for identity validation

3. FUNCTIONAL REQUIREMENTS

Each functional requirement is assigned a unique identifier in the format FR-[Subsystem Code]-[Number]. Priority is classified as MUST (critical for Phase 1), SHOULD (important but deferrable to Phase 1.x), or COULD (Phase 2+).

3.1 Subsystem 1: Identity and Trust (IT)

3.1.1 Authentication

Req. ID	Requirement Description	Priority
FR-IT-001	The system SHALL authenticate all users exclusively through UTM's OAuth2 SSO endpoint. No external email/password registration is permitted.	MUST
FR-IT-002	Upon successful OAuth2 authentication, the system SHALL retrieve and store the user's matriculation number, full name, role (student/staff), faculty, and college affiliation from the UTM identity provider.	MUST
FR-IT-003	The system SHALL generate and maintain a signed JWT (JSON Web Token) for each authenticated session with an expiry of 24 hours.	MUST
FR-IT-004	The system SHALL automatically create a new user profile in Firestore upon the first successful login, linking the profile to the UTM matriculation number as the primary key.	MUST
FR-IT-005	The system SHALL prevent any user action (browsing, listing, requesting) without an active authenticated session.	MUST
FR-IT-006	The system SHALL allow users to log out, which invalidates the JWT and clears the local session cache.	MUST

3.1.2 User Profile

Req. ID	Requirement Description	Priority
FR-IT-007	The system SHALL display a user profile page showing: display name, college, faculty, member since date, Trust Rating, total items lent, total items borrowed, and current Good Standing status.	MUST
FR-IT-008	Users SHALL be able to upload a profile photo to their account.	SHOULD
FR-IT-009	The system SHALL calculate and display a Trust Rating (0.0–5.0) based on completed transaction ratings received from other users.	MUST
FR-IT-010	The system SHALL define Good Standing as a Trust Rating >= 3.0 and zero unresolved penalty flags. Only users in Good Standing may initiate new borrow requests.	MUST
FR-IT-011	Users SHALL be able to view the public profile of any other UTM Borrow user (excluding private contact details).	MUST

3.2 Subsystem 2: Distributed Inventory (DI)

3.2.1 Item Listing Creation

Req. ID	Requirement Description	Priority
FR-DI-001	Authenticated users SHALL be able to create a new item listing by providing: item name, category, description, condition rating (New/Good/Fair/Poor), and minimum 1 high-resolution photo.	MUST
FR-DI-002	The system SHALL support the following item categories: Lab Equipment, Books & References, Stationery & Drawing Tools, Electronics, Workshop Tools, Clothing & PPE, and Others.	MUST
FR-DI-003	Users SHALL be able to specify a maximum borrow duration (in days) and maximum concurrent loans for each listed item.	MUST
FR-DI-004	Users SHALL be able to set item visibility to Public (all UTM Borrow users) or Faculty-Only (restricted to users in the same faculty).	SHOULD
FR-DI-005	The system SHALL associate each listing with the lender's college geolocation for proximity-based search filtering.	MUST
FR-DI-006	Each listing SHALL be assigned one of three availability states: Available, Pending (request under review), or In Use (currently borrowed).	MUST
FR-DI-007	Users SHALL be able to edit any field of their own listing while it is in Available state. Editing is locked during Pending and In Use states.	MUST
FR-DI-008	Users SHALL be able to deactivate (soft-delete) their listing at any time when it is in Available state. Deactivated listings are hidden from search but retained in the database.	MUST

3.2.2 Search and Discovery

Req. ID	Requirement Description	Priority
FR-DI-009	The system SHALL provide a full-text search bar that queries item names and descriptions across all active listings.	MUST
FR-DI-010	Search results SHALL be filterable by: category, item condition, proximity (college distance), and availability status.	MUST
FR-DI-011	Search results SHALL default to sorting by proximity (nearest college first) when the user's college affiliation is known.	MUST
FR-DI-012	The system SHALL display only listings in Available state in default search results. Users may optionally include In Use listings to see future availability.	MUST
FR-DI-013	Each search result card SHALL display: item photo, item name, category, condition, lender's college, and lender's Trust Rating.	MUST

3.3 Subsystem 3: Borrow Request Flow (BR)

Req. ID	Requirement Description	Priority

FR-BR-001	A user in Good Standing SHALL be able to send a borrow request for any Available item by specifying a requested borrow duration (within the lender's maximum).	MUST
FR-BR-002	The system SHALL prevent a user from sending a borrow request for their own listed item.	MUST
FR-BR-003	The system SHALL send a real-time push notification and in-app notification to the Lender when a new request is received.	MUST
FR-BR-004	The Lender SHALL be able to Approve or Reject a request from their notification or dashboard. Approval transitions the listing to Pending state.	MUST
FR-BR-005	The system SHALL notify the Borrower via push notification upon Approval or Rejection of their request.	MUST
FR-BR-006	Upon approval, the system SHALL calculate the return deadline as: approval date + agreed borrow duration, and present this deadline to both parties.	MUST
FR-BR-007	A Lender SHALL be able to cancel an approved request (before handover QR scan) with a mandatory reason. The listing returns to Available state.	MUST
FR-BR-008	A Borrower SHALL be able to cancel their own request while it is in Pending approval state.	MUST
FR-BR-009	The system SHALL maintain a full request history (including rejected and cancelled requests) in each user's transaction log.	MUST

3.4 Subsystem 4: QR Handover & State Management (QH)

3.4.1 Item Dispatch (Lender to Borrower)

Req. ID	Requirement Description	Priority
FR-QH-001	Upon borrow approval, the Lender SHALL be able to generate a Handover QR code from their active transaction dashboard.	MUST
FR-QH-002	The Handover QR code SHALL encode a cryptographically signed, time-limited session token (60-second expiry) containing: transaction ID, lender ID, borrower ID, and timestamp.	MUST
FR-QH-003	The Borrower SHALL use the UTM Borrow in-app camera scanner to scan the Handover QR code. Screenshots or external QR readers SHALL not be accepted.	MUST
FR-QH-004	Upon successful dual-scan verification, the system SHALL transition the item state from Pending to In Use and record the exact handover timestamp.	MUST
FR-QH-005	If the QR token expires before scanning, the Lender SHALL regenerate a new token. Expired tokens are automatically invalidated.	MUST
FR-QH-006	The system SHALL require the Lender to upload a Pre-Borrow Photo at the moment of Handover QR generation as a condition record baseline.	MUST

3.4.2 Item Return (Borrower to Lender)

Req. ID	Requirement Description	Priority
FR-QH-007	The Return process SHALL mirror the Dispatch process: the Borrower generates a Return QR code, and the Lender scans it to confirm receipt.	MUST
FR-QH-008	Upon Return QR scan, the system SHALL prompt both users to complete a Condition Check: the Borrower uploads a Post-Borrow Photo, and the Lender marks condition as Unchanged, Minor Damage, or Major Damage.	MUST
FR-QH-009	If condition is marked as Minor Damage or Major Damage, the system SHALL initiate a Condition Dispute workflow and freeze the Borrower's Borrowing Rights pending resolution.	MUST
FR-QH-010	Upon successful return with Unchanged condition, the system SHALL transition the item to Available, release the Borrower's Borrowing Rights, and prompt both parties to submit a transaction rating.	MUST

3.5 Subsystem 5: Lease Cycle Management (LC)

Req. ID	Requirement Description	Priority
FR-LC-001	The system SHALL maintain a Lease Cycle State for every transaction with the following states: Request Sent, Request Approved, Dispatched (In Use), Return Initiated, Returned, Disputed, Overdue, Closed.	MUST
FR-LC-002	A Cloud Function SHALL automatically transition an In Use transaction to Overdue state if the return deadline passes without a Return QR scan.	MUST
FR-LC-003	The system SHALL send a reminder notification to the Borrower 48 hours before the return deadline.	MUST
FR-LC-004	The system SHALL send a final reminder notification to the Borrower 24 hours before the return deadline.	MUST
FR-LC-005	When a transaction enters Overdue state, the Borrower's account SHALL be flagged and Borrowing Rights suspended until the item is returned or the dispute is resolved.	MUST
FR-LC-006	The Lender SHALL be able to submit an Overdue Report which escalates the transaction to an Admin for review and generates a preliminary report.	MUST

3.6 Subsystem 6: Incentive, Feedback & Penalty (IF)

Req. ID	Requirement Description	Priority
FR-IF-001	Upon successful Lease Cycle completion, both Lender and Borrower SHALL be prompted to submit a 1–5 star rating for the transaction.	MUST
FR-IF-002	The system SHALL calculate and update each user's aggregate Trust Rating as a weighted average of all received ratings.	MUST
FR-IF-003	Users with a Trust Rating below 3.0 SHALL automatically lose Good Standing status and have Borrowing Rights suspended until the rating improves.	MUST

FR-IF-004	Administrators SHALL be able to manually suspend, warn, or reinstate any user account from the Admin Panel.	MUST
FR-IF-005	The system SHALL award Merit Points for completed lending transactions based on item category and number of borrows facilitated (Phase 2).	COULD
FR-IF-006	The system SHALL support a 1 RM maintenance fee per transaction through a digital wallet system, deducted at time of borrow approval (Phase 2).	COULD
FR-IF-007	The system SHALL display a leaderboard of top community contributors (highest Merit Points) visible to all users (Phase 2).	COULD

3.7 Subsystem 7: Notification Engine (NE)

Req. ID	Requirement Description	Priority
FR-NE-001	The system SHALL support push notifications to mobile devices via Firebase Cloud Messaging (FCM).	MUST
FR-NE-002	The system SHALL support in-app notification centre displaying all system alerts with read/unread status.	MUST
FR-NE-003	Users SHALL be able to configure notification preferences (enable/disable push notifications per event type).	SHOULD
FR-NE-004	All transactional notifications (request approval, QR handover, return deadline) SHALL also be sent to the user's UTM email address as a secondary channel.	SHOULD
FR-NE-005	System notifications SHALL be sent in the user's preferred language (Bahasa Malaysia or English).	SHOULD

3.8 Subsystem 8: Admin Moderation Panel (AM)

Req. ID	Requirement Description	Priority
FR-AM-001	The Admin Panel SHALL be accessible only to users with the Administrator role, enforced via RBAC.	MUST
FR-AM-002	Administrators SHALL be able to view all user accounts, transaction logs, active disputes, and overdue reports.	MUST
FR-AM-003	Administrators SHALL be able to suspend, warn, or reinstate any user account with a mandatory reason and audit trail.	MUST
FR-AM-004	Administrators SHALL be able to export any transaction log as a PDF report including timestamps, user details, item details, and state history.	MUST
FR-AM-005	Administrators SHALL be able to mark a dispute as Resolved in favour of Lender or Borrower, which automatically updates the relevant user accounts.	MUST
FR-AM-006	The Admin Panel SHALL display a real-time dashboard showing: total active listings, active borrow transactions, overdue count, dispute queue count, and registered users.	MUST

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

Req. ID	Requirement Description	Target Metric
NFR-PE-001	Search query results SHALL be returned within acceptable time under normal campus network conditions.	< 2 seconds (95th percentile)
NFR-PE-002	QR code generation SHALL be near-instantaneous to support in-person handover flow.	< 500 milliseconds
NFR-PE-003	Push notifications SHALL be delivered to the target device promptly after triggering event.	< 5 seconds (P95)
NFR-PE-004	The system SHALL support concurrent active users without performance degradation.	Up to 500 concurrent users
NFR-PE-005	Firebase Firestore read/write operations SHALL complete within acceptable response times.	< 1 second for 95% of operations

4.2 Security Requirements

Req. ID	Requirement Description	Priority
NFR-SE-001	All API endpoints SHALL require a valid JWT in the Authorization header. Unauthenticated requests SHALL return HTTP 401.	MUST
NFR-SE-002	The system SHALL implement Role-Based Access Control (RBAC) ensuring users can only access and modify their own data, except Admin roles.	MUST
NFR-SE-003	All data transmission between client and server SHALL use TLS 1.2 or higher (HTTPS only). HTTP requests SHALL be redirected.	MUST
NFR-SE-004	QR handover tokens SHALL be cryptographically signed (HMAC-SHA256 minimum) and expire after 60 seconds to prevent replay attacks.	MUST
NFR-SE-005	Item photos and user data SHALL be stored in Firebase Cloud Storage with access rules preventing unauthorized cross-user access.	MUST
NFR-SE-006	The system SHALL log all authentication attempts, QR scan events, and admin actions with timestamps for audit purposes.	MUST
NFR-SE-007	The system SHALL comply with Malaysia's Personal Data Protection Act 2010 (PDPA). User data SHALL not be shared with third parties without consent.	MUST
NFR-SE-008	The system SHALL implement input validation and sanitization on all user-submitted text fields to prevent XSS and injection attacks.	MUST

4.3 Reliability and Availability Requirements

Req. ID	Requirement Description	Target Metric
NFR-RA-001	The platform SHALL maintain high availability during core campus hours.	99.5% uptime (Mon–Fri, 7am–11pm)
NFR-RA-002	Firestore transaction writes SHALL be atomic, ensuring no partial state updates occur during Lease Cycle transitions.	100% atomicity guaranteed
NFR-RA-003	Firebase Cloud Functions SHALL implement retry logic for failed notification delivery attempts.	3 retry attempts with exponential backoff
NFR-RA-004	In the event of a backend service outage, the frontend SHALL display a clear maintenance message and not silently fail.	Graceful degradation required

4.4 Usability Requirements

Req. ID	Requirement Description	Priority
NFR-US-001	A new user SHALL be able to complete registration, create a listing, and send a borrow request within their first session without external help.	MUST
NFR-US-002	The mobile application SHALL support both portrait and landscape orientations on standard smartphone screen sizes (5 inch to 6.9 inch).	MUST
NFR-US-003	All user interface text SHALL be available in Bahasa Malaysia and English, switchable from user settings.	MUST
NFR-US-004	The QR handover interface SHALL include clear, illustrated step-by-step instructions visible at point of use.	MUST
NFR-US-005	The application SHALL conform to WCAG 2.1 Level AA accessibility standards for color contrast and touch target sizes.	SHOULD
NFR-US-006	Error messages SHALL clearly explain what went wrong and provide an actionable resolution step.	MUST

4.5 Maintainability Requirements

- The backend API SHALL be structured as modular, independently deployable Node.js service modules (one module per subsystem)
- All Firestore collections SHALL follow a consistent, documented schema with version fields to support future migrations
- The codebase SHALL maintain a minimum of 70% unit test coverage on all business logic functions
- All Cloud Functions SHALL be documented with inline JSDoc annotations describing inputs, outputs, and side effects
- The application SHALL support feature-flag-based rollout of Phase 2 features without a full redeployment

4.6 Scalability Requirements

- The Firestore data model SHALL be designed to avoid hot-spots under high concurrent read/write load (no single shared counter documents)
- Firebase Cloud Functions SHALL scale automatically to handle sudden spikes (e.g., semester start/end)
- The search query architecture SHALL support paginated results (20 items per page) to prevent excessive data transfer on mobile networks
- The system SHALL be architecturally ready to support multiple UTM campuses (e.g., UTM KL) with campus-scoped data partitioning in Phase 3

5. SYSTEM ARCHITECTURE & DATA MODEL

5.1 High-Level Architecture

UTM Borrow follows a three-tier client-server architecture with a serverless backend pattern:

- Tier 1 — Presentation Layer: React Native mobile app (iOS/Android) and React.js progressive web app (PWA). Communicates with the backend exclusively via REST API over HTTPS.
- Tier 2 — Application Layer: Node.js REST API deployed on Firebase Cloud Functions. Implements all business logic, authentication verification, and state machine transitions. Organized into eight subsystem modules corresponding to the functional requirements.
- Tier 3 — Data Layer: Google Firebase Firestore (primary database), Firebase Cloud Storage (media files), Firebase Cloud Messaging (push notifications), and UTM OAuth2 SSO (identity provider — external dependency).

5.2 Core Firestore Data Collections

Collection	Primary Fields	Relationships
users	uid (UTM matric), name, role, faculty, college, trustRating, goodStanding, meritPoints, createdAt	1-to-many with listings, transactions
listings	listingId, ownerId, title, category, condition, photos[], status, maxDuration, visibility, collegeGeo, createdAt	Many-to-1 with users; 1-to-many with transactions
transactions	txId, listingId, lenderId, borrowerId, status (Lease Cycle State), requestDate, approvalDate, dispatchDate, returnDeadline, returnDate, conditionResult	Joins users and listings; 1-to-1 with qrTokens
qrTokens	tokenId, txId, type (dispatch/return), signedToken, expiresAt, usedAt, isValid	1-to-1 with transactions
ratings	ratingId, txId, fromUserId, toUserId, score, comment, createdAt	Many-to-1 with transactions and users
disputes	disputeId, txId, reportedBy, reason, prePhoto, postPhoto, status, adminId, resolution, createdAt	1-to-1 with transactions
notifications	notifId, userId, type, title, body, isRead, relatedTxId, createdAt	Many-to-1 with users
auditLogs	logId, actorId, action, targetType, targetId, metadata, timestamp	Reference to users and any entity

5.3 Lease Cycle State Machine

The following state transitions define the complete lifecycle of a borrow transaction. All transitions are enforced atomically by the application layer and cannot be skipped or reversed except by Administrator override:

Current State	Trigger Event	Next State	Actor
(none)	Borrower sends request	Request Sent	Borrower
Request Sent	Lender approves	Request Approved	Lender
Request Sent	Lender rejects / Borrower cancels	Closed (Rejected)	Lender / Borrower
Request Approved	Lender generates & Borrower scans Dispatch QR	Dispatched (In Use)	Both (QR System)
Request Approved	Lender cancels before handover	Closed (Cancelled)	Lender
Dispatched (In Use)	Return deadline passes without return	Overdue	System (Auto)
Dispatched (In Use)	Borrower generates & Lender scans Return QR	Return Initiated	Both (QR System)
Return Initiated	Lender marks condition Unchanged	Returned (Closed)	Lender
Return Initiated	Lender marks condition Damaged	Disputed	Lender
Overdue	Borrower returns item (late)	Return Initiated	Both (QR System)
Overdue	Lender submits Overdue Report	Escalated to Admin	Lender
Disputed	Admin resolves in Lender favour	Closed (Penalty Applied)	Administrator
Disputed	Admin resolves in Borrower favour	Closed (Cleared)	Administrator

5.4 API Endpoint Summary

Method	Endpoint	Description	Auth Required
POST	/auth/callback	UTM OAuth2 callback handler — creates/updates user profile and issues JWT	No
GET	/users/:uid	Retrieve public user profile	Yes
GET	/listings	Search listings with filters and pagination	Yes
POST	/listings	Create a new item listing	Yes
PUT	/listings/:id	Update an existing listing (owner only)	Yes
DELETE	/listings/:id	Deactivate a listing (soft delete)	Yes
POST	/transactions	Submit a borrow request	Yes
PUT	/transactions/:id/approve	Approve a borrow request (lender only)	Yes

PUT	/transactions/:id/reject	Reject a borrow request (lender only)	Yes
POST	/transactions/:id/qr/dispatch	Generate Dispatch QR token	Yes
POST	/transactions/:id/qr/scan	Validate scanned QR token and transition state	Yes
POST	/transactions/:id/qr/return	Generate Return QR token	Yes
POST	/transactions/:id/condition	Submit post-return condition report and photos	Yes
POST	/transactions/:id/rating	Submit transaction rating	Yes
POST	/disputes	Create a damage dispute	Yes
PUT	/disputes/:id/resolve	Resolve a dispute (admin only)	Yes (Admin)
GET	/admin/dashboard	Retrieve admin dashboard metrics	Yes (Admin)
GET	/admin/users	List all users with filters	Yes (Admin)
PUT	/admin/users/:uid/suspend	Suspend a user account	Yes (Admin)
GET	/admin/transactions/:id/export	Export transaction log as PDF	Yes (Admin)

6. USE CASE SPECIFICATIONS

UC-01: Happy Path — Successful Item Borrow and Return

Field	Detail
Use Case ID	UC-01
Use Case Name	Complete Successful Borrow Transaction (Happy Path)
Primary Actor	Borrower (Student)
Secondary Actor	Lender (Student), System (Cloud Functions)
Preconditions	Both users are authenticated. Borrower has Good Standing status. Item is in Available state.
Postconditions	Item returns to Available state. Both users receive updated Trust Ratings. Transaction logged as Closed.
Main Success Scenario	1. Borrower searches for item and selects available listing. 2. Borrower submits borrow request with desired duration. 3. System sends push notification to Lender. 4. Lender approves request; system calculates return deadline. 5. Both users receive approval notification with deadline. 6. Lender generates Dispatch QR; uploads Pre-Borrow Photo. 7. Borrower scans Dispatch QR via in-app camera. 8. System transitions item to In Use; records dispatch timestamp. 9. System sends 48hr and 24hr deadline reminders to Borrower. 10. Borrower generates Return QR on/before deadline. 11. Lender scans Return QR; marks condition as Unchanged. 12. System transitions item to Available; prompts both users for rating. 13. Both users submit ratings; Trust Ratings updated. 14. Transaction logged as Closed.
Alternative Flow A	Step 4: Lender rejects request. Borrower notified. Transaction closed. Item remains Available.
Alternative Flow B	Step 11: Lender marks condition as Damaged. Dispute workflow initiated (see UC-03).
Exception Flow	Steps 6-7: QR token expires before scan. Lender regenerates new 60-second token.

UC-02: Overdue Item and Escalation

Field	Detail
Use Case ID	UC-02
Use Case Name	Overdue Item Handling and Admin Escalation
Primary Actor	System (Cloud Function)
Secondary Actor	Borrower, Lender, Administrator
Preconditions	Transaction is in Dispatched (In Use) state. Return deadline has passed without Return QR scan.
Postconditions	Borrower's account is flagged Overdue. Borrowing Rights suspended.
Main Success Scenario	1. System detects return deadline passed at midnight. 2. System auto-transitions transaction to Overdue state. 3. System sends Overdue alert

	to Borrower and Lender. 4. Borrower's Borrowing Rights suspended. 5. Lender chooses to wait or submit Overdue Report. 6. If Overdue Report submitted, Admin receives notification. 7. Admin reviews transaction log and contacts student via UTM records. 8. Admin resolves case — item returned or penalty applied. 9. Admin closes transaction; Borrower Rights reinstated or permanently restricted.
Alternative Flow	Step 5: Borrower returns item late. Return QR process completes. Account flagged with Overdue history but Rights reinstated after return.

UC-03: Damage Dispute Resolution

Field	Detail
Use Case ID	UC-03
Use Case Name	Item Condition Dispute Workflow
Primary Actor	Lender
Secondary Actor	Borrower, Administrator
Preconditions	Transaction is in Return Initiated state. Lender has scanned Return QR. Lender marks condition as Damaged.
Postconditions	Dispute resolved by Administrator. Appropriate penalty or clearance applied to Borrower.
Main Success Scenario	1. Lender marks condition as Minor/Major Damage during Return QR scan. 2. System prompts Borrower to upload Post-Borrow Photo. 3. System creates Dispute record linking Pre- and Post-Borrow Photos. 4. System freezes Borrower's Borrowing Rights. 5. Admin receives Dispute queue notification. 6. Admin reviews photo comparison, transaction log, and both user statements. 7. Admin rules in favour of Lender (damage confirmed). 8. System applies penalty to Borrower (Trust Rating reduction, replacement fee trigger). 9. Admin marks Dispute Resolved. Transaction logged as Closed (Penalty Applied). 10. Both users notified of resolution.
Alternative Flow	Step 7: Admin rules in favour of Borrower (no damage detected). Borrower Rights restored. Lender rating unaffected. Transaction Closed (Cleared).

7. APPENDICES

Appendix A: Revision History

Version	Date	Author	Description of Change
0.1	March 2026	Development Team	Initial draft — scope and subsystem identification
0.5	April 2026	Development Team	Full functional requirements first draft
1.0	April 2026	Development Team + Supervisor Review	Final approved version — all NFRs, use cases, and data model included

Appendix B: Stakeholder Sign-Off

Role	Name	Signature	Date
Project Supervisor / Dr.		_____	_____
Lead Developer		_____	_____
UTM IT Department Representative		_____	_____
UTMCS Representative		_____	_____

Appendix C: Technology Stack Summary

Layer	Technology	Justification
Mobile Frontend	React Native	Cross-platform iOS/Android from single codebase; large UTM student developer community familiarity
Web Frontend	React.js (PWA)	Progressive Web App for browser access without app installation; same component library as mobile
Backend API	Node.js + Firebase Cloud Functions	Serverless scaling; native Firebase SDK integration; cost-effective for variable campus traffic
Database	Firebase Firestore	Real-time sync; NoSQL flexibility for evolving item/transaction schema; strong offline support
Authentication	UTM OAuth2 SSO + Firebase Auth	Closes the ecosystem to UTM users only; eliminates fake account creation
Media Storage	Firebase Cloud Storage	Secure, scalable photo storage with RBAC; direct mobile SDK upload support
Push Notifications	Firebase Cloud Messaging (FCM)	Zero-cost for required message volume; native iOS and Android integration
QR Token Security	HMAC-SHA256 signed JWTs	Cryptographic tamper protection; 60-second expiry enforced server-side

Appendix D: Economic Impact Projection

The following projections are based on a conservative adoption model assuming 10% of the 20,000 UTM student population actively uses the platform within the first academic year:

Metric	Assumption	Projected Value
Active Users (Year 1)	10% of 20,000 students	~2,000 users
Average item value listed	Mix of books (RM 60–200), tools (RM 100–400)	~RM 150 average
Transactions per user per semester	Conservative: 2 borrows/semester	~4,000 transactions/year
Total student savings (Year 1)	4,000 tx x RM 150 avg value avoided	~RM 600,000 saved
Items diverted from purchase/waste	4,000 transactions = 4,000 fewer purchases	~2–4 tonnes material waste avoided
Server costs (Firebase Spark/Blaze)	Blaze pay-as-you-go estimated	~RM 200–500/month at scale
Phase 2: Fee revenue (1 RM/tx)	If 4,000 annual transactions x RM 1	~RM 4,000/year (self-sustaining)

END OF DOCUMENT

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